

Basic Principles of Medicine 1
Module : Foundations to Medicine
FTM 11

Lecture Title
**Genetic Variation, Genes and
Chromosomes**

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Recommended Reading

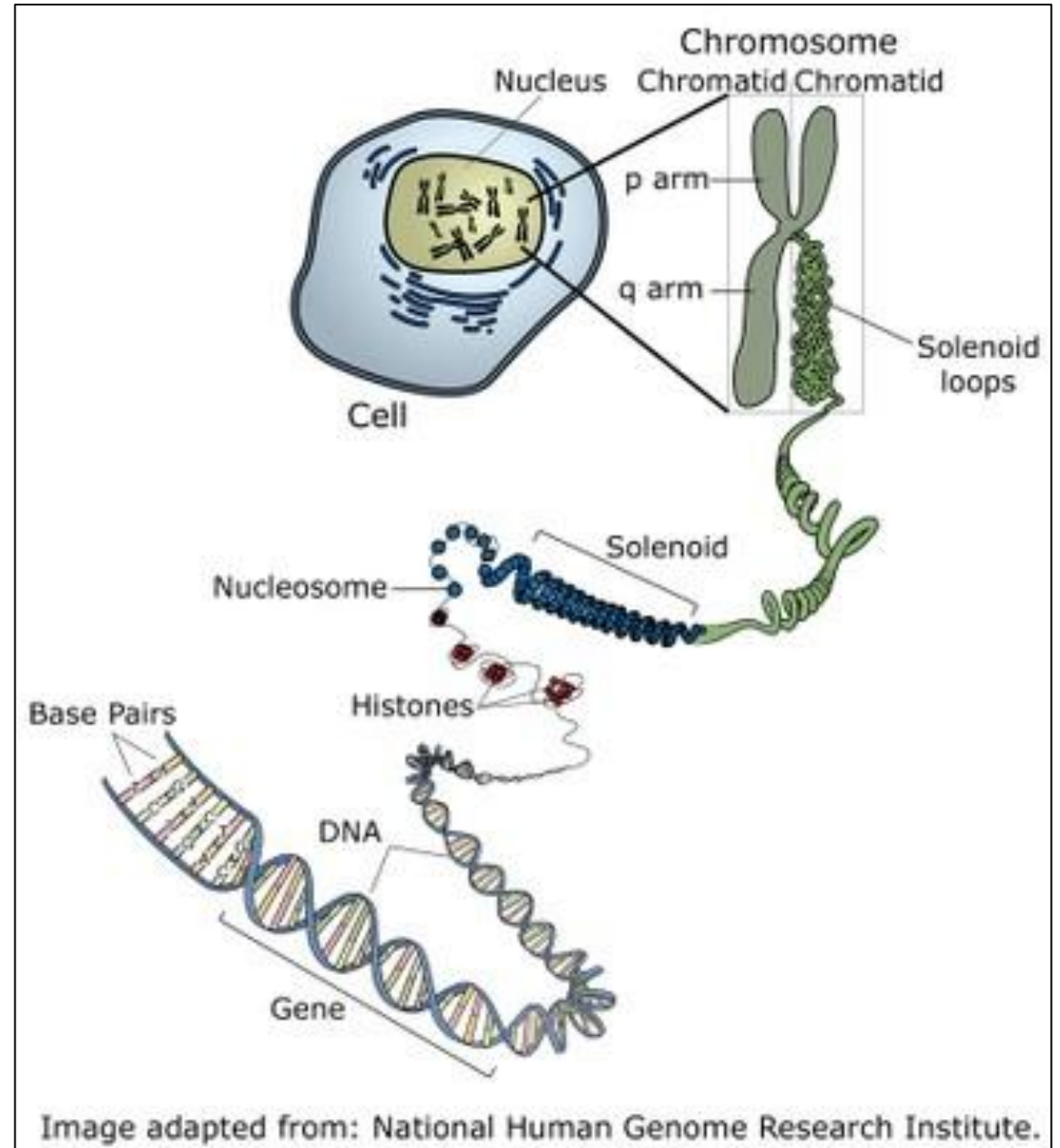
Korf & Irons: Sources of Information 1.1 – OMIM – p4
Self-Assessment Review Questions: Korf & Irons

Objectives

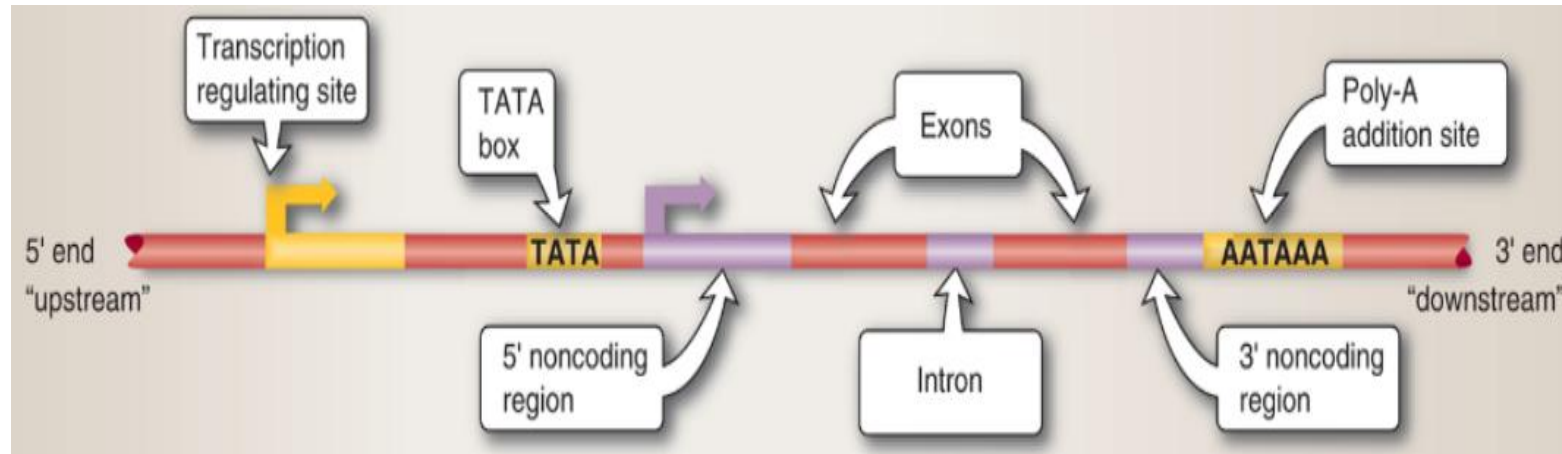
SOM.MKIII.BPM1.1.FTM.3.GNET.0016	Compare and contrast different types of repetitive DNA including low copy repeats, SSR, VNTR, LINE, SINE, pseudogenes, and gene families
SOM.MKIII.BPM1.1.FTM.3.GNET.0017	Compare and contrast single nucleotide polymorphisms to rare variants
SOM.MKIII.BPM1.1.FTM.3.GNET.0018	Discuss the structure of chromosomes in the human (male and female), and general terminology used to describe karyotypes; Compare nuclear DNA to organellar DNA (such as the mitochondrial)
SOM.MKIII.BPM1.1.FTM.3.GNET.0019	Analyze dosage compensation by X-chromosome inactivation
SOM.MKIII.BPM1.1.FTM.3.GNET.0105	Explain the normal function of epigenetic gene silencing in development; including the role of X-inactivation

DNA – Basis of Heredity

DNA
to
Genes
to
Chromosomes

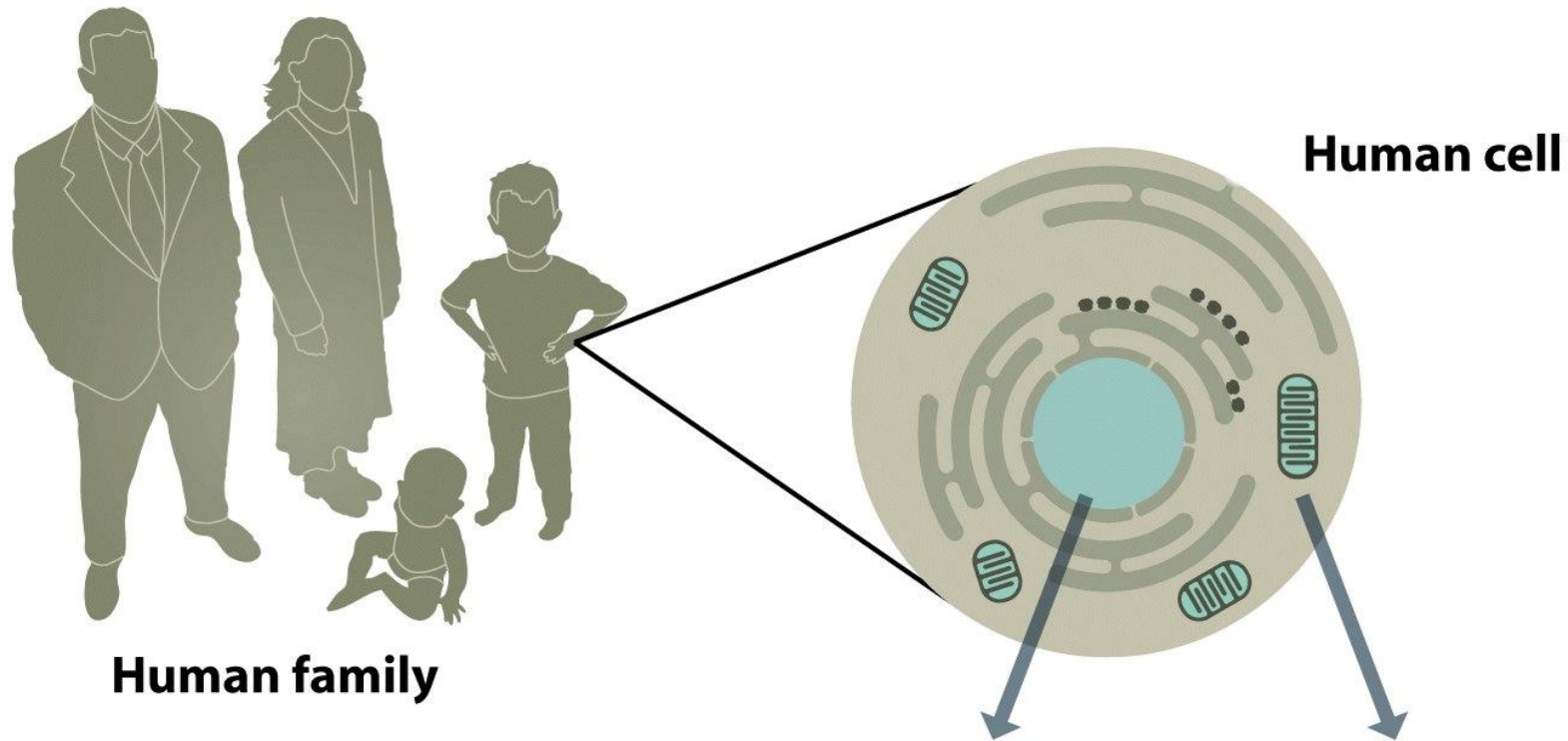


Eukaryotic Gene Structure



- coding regions (exons), noncoding regions (introns) and regulatory sequences (promoters, enhancers, silencers)
- The sequence of the **exon and introns are transcribed**. The introns are removed and the exons are joined by **SPLICING**. Only the exon sequences are translated to form proteins.
- mRNA may undergo '**alternate splicing**' or **RNA editing** to result in the formation of different types of proteins
- The regulatory sequences regulate the rates at which transcription of the gene occurs
- Epigenetic modifications such as methylation may regulate the rate at which transcription occurs

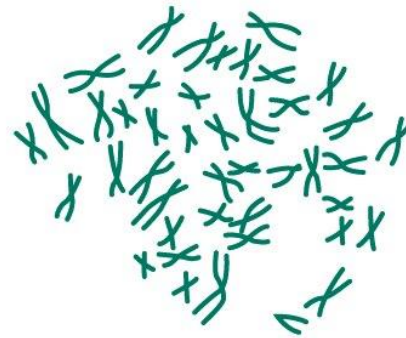
How much DNA do we have, and what is it made up of?



Nuclear genome

Mitochondrial genome

- About 3 billion bp (haploid), over 23 chromosomes
- This is huge

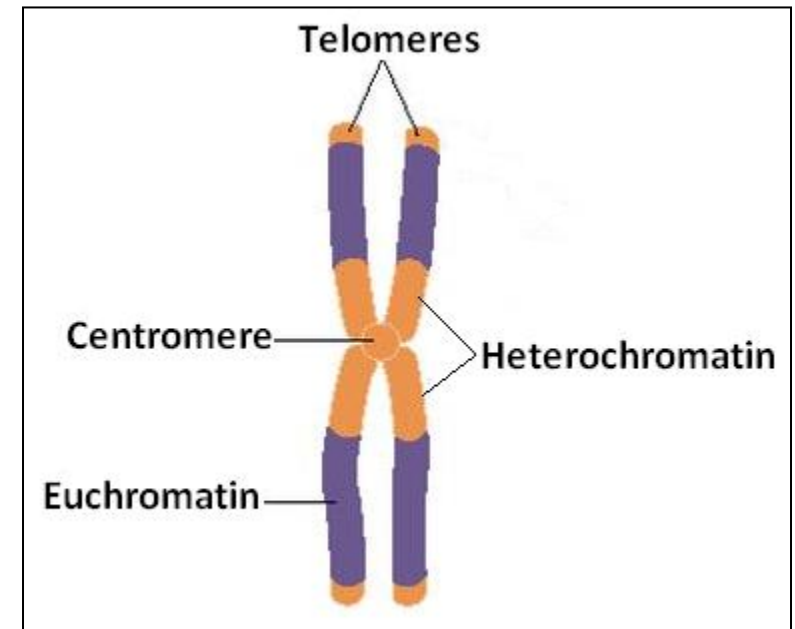
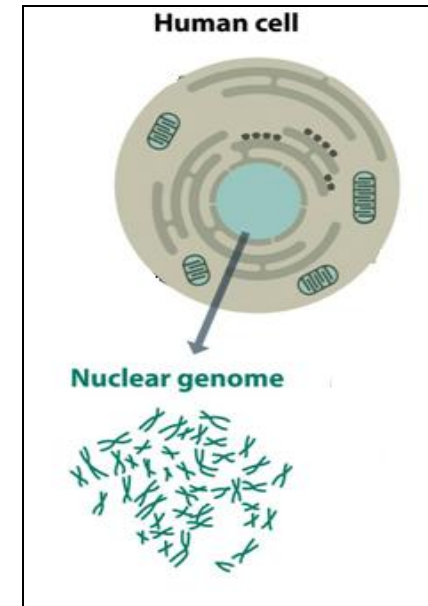


- About 16.7 kb (circular)
- In comparison to nuclear genome, this is miniscule

SOM.1a.BPM1.1.FTM.3.GNET.0014

Nuclear Genes

- There are about 20-25,000 unique copy genes in the nuclear genome.
- High gene density in euchromatin.
- Heterochromatin and centromeric regions are generally non-coding.
- Genes are variable in size
 - By far the largest human gene is the dystrophin gene ~2.5Mb – 2.7 Mb depending on how it is defined.
 - Other genes may be very (very) small.

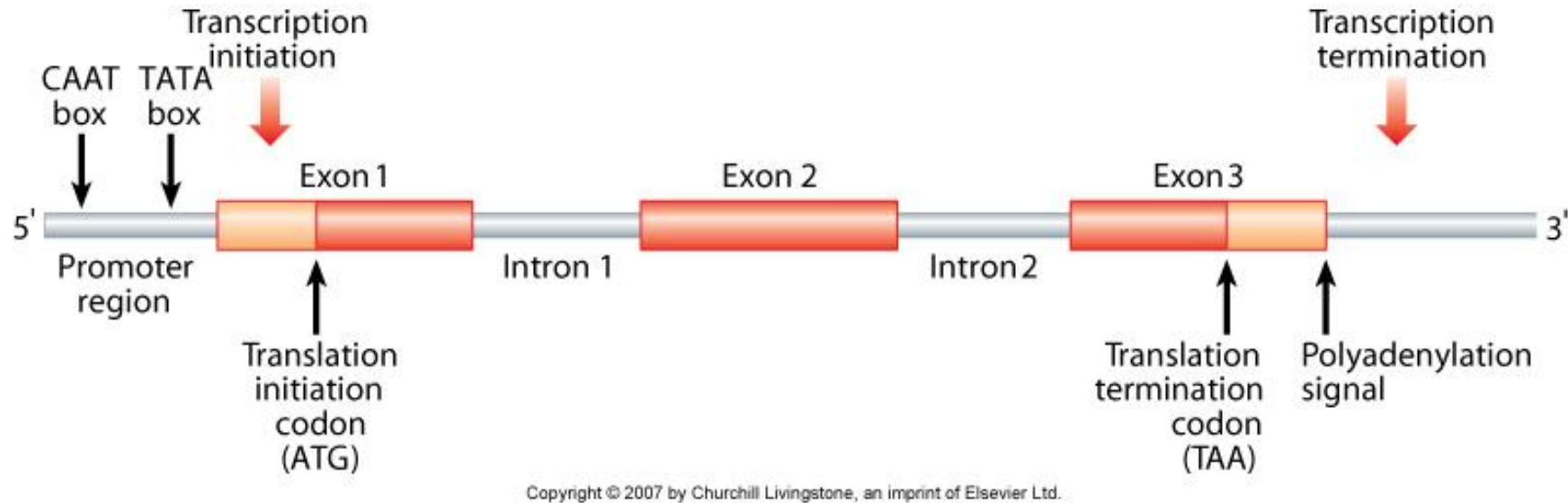


Chromosome

Nuclear Coding DNA

(Unique Single Copy Genes and Multigene Families)

Unique single copy genes



1) Single copy genes

- Unique sequences in the genome that code for a protein:
- receptors, enzymes, hormones, structural cellular elements, etc

2) Multigene families (can be clustered together or dispersed in different genomic locations): Genes with similar functions that have arisen by gene duplication

- **Classic gene families**

- multicopy genes that show a high degree of homology (*e.g.* HOX genes, globin genes, and genes for rRNAs, tRNAs)

- **Gene superfamilies**

- multicopy genes with similar function but limited gene homology (*e.g.* HLA genes, T-cell receptors)

Extragenic DNA

Less than 2% of our genomic DNA actually encodes protein. Most DNA is apparently not transcriptionally active and often appears to have no function. In the past this has been called “Junk DNA”, but maybe it has a function.... Maybe regulatory, maybe something we haven't thought of yet?

- Extragenic DNA constitutes the majority of the human genome
- Extragenic DNA is predominantly transcriptionally inactive
- Extragenic DNA might play a **role in regulation** of gene expression
- Tandem repeated DNA sequences consist of blocks of tandem repeats of non-coding DNA. (length of repeat sequence determines how it is named)
- The shorter the repeat length, the more **polymorphic that sequence is**
- Tandem repeats **are inherited in a co-dominant fashion; one from each parent**
- Simple sequence repeat variations are used as the basis of DNA fingerprinting

Some types of DNA variation

- **SNP**- Single nucleotide polymorphism
 - Formally defined as a variant that is found at least in 1% of the population.
 - Single base pair change between individuals
- **SSR** (Simple Sequence Repeat) – these is the most simple type of repetitive sequence; and most polymorphic
 - These are tandem repeats of 2, 3 or 4 bp, repeated many times
 - Some texts will call these simple tandem repeats (STR), but this may lead to confusion with VNTR so we won't use that terminology
- **VNTR** (sometimes called short tandem repeats, but this is confusing to SSR),
 - These repeats are a bit longer then the SSR, maybe 5 bp, 10 bp, or hundred of bp; repeating many times
- **LCR** (low copy repeat) – may be very large

See page 28 -31, and p82 in Korf textbook; note that in the literature, and in almost *all* textbook sources, these are sometimes defined and named differently – but idea is always the same – we present what we think are the most consistent definitions.

SNP – the **s**ingle **n**ucleotide **p**olymorphism

- Human genome is $\sim 3 \times 10^9$ bp for haploid
- Approximately 1 in each thousand bp there is a common variation (common means that $>1\%$ of the population has the change; polymorphism)
- Therefore there are about 3 million common single nucleotide bp changes – called SNP.
- Most SNP do not have an associated phenotype



- Birth of “*recreational genomics*” as direct to consumer (DTC) companies will now provide SNP maps of **your** genome using array based technology. (see chapter 14 of Korf text); spit + 199 \$US.

- We come back to this many times in this course

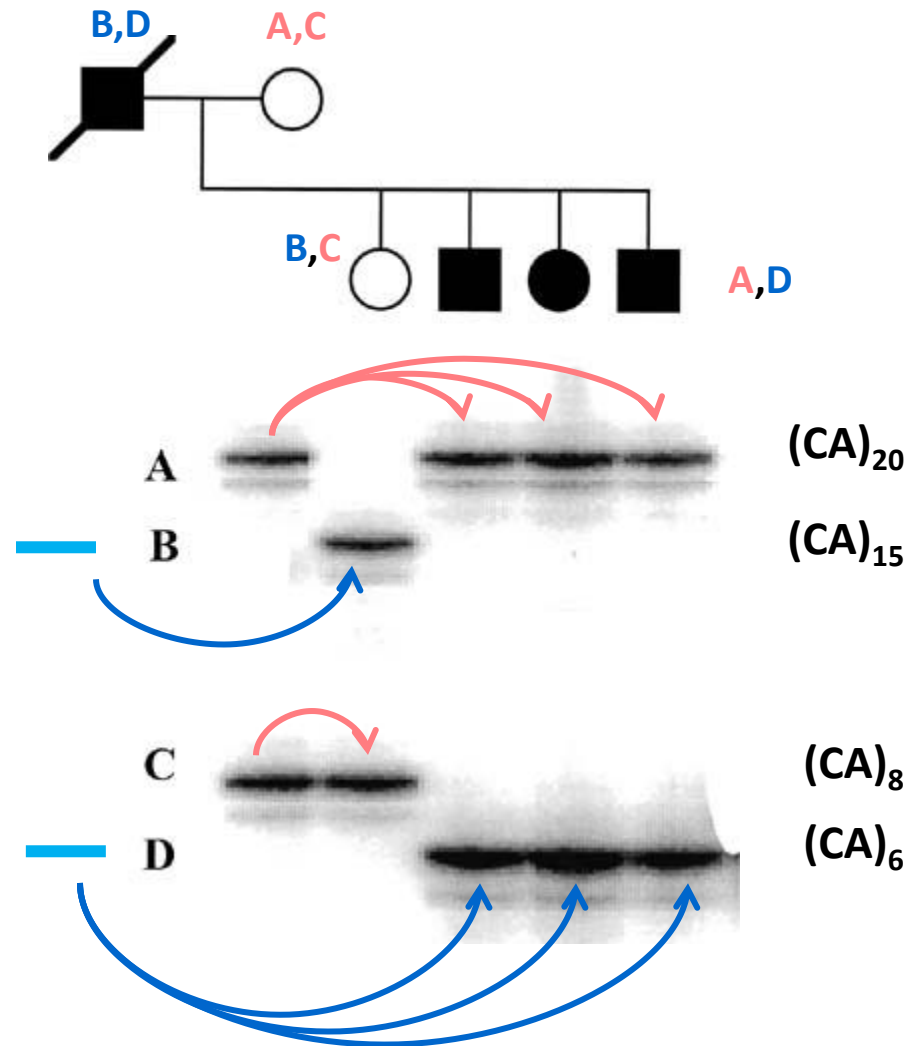
Application of SSR analysis of a CA repeat linked to an autosomal-dominant disease

- This is tracking an SSR, since it is a variation in tandem repeats of a dinucleotide sequence

- Dad's genotype is most likely B,D. the 6 tandem CA repeat 'allele D' is most likely linked to the disease in his family

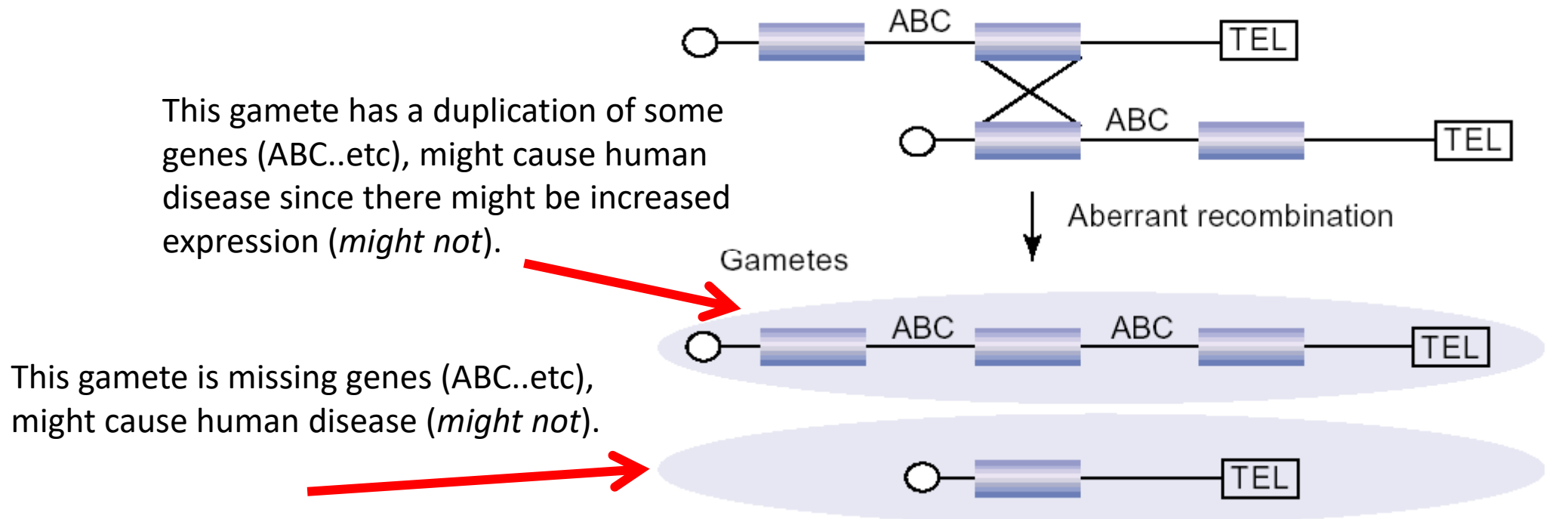
CA repeat 'allele D'
linked to the disease

- We will spend many hours (later, next week) learning how to analyze and interpret this type of data
- SSR are very useful in forensics and paternity testing.
- Easy to detect by PCR based methods



The low copy repeat

- These may be thousands (kb) to many hundreds of thousands of bp long (that is, they can be big)
- May be repeated just twice in the genome, or many times
- May cause mispairing during meiosis or mitosis



Highly repeated Interspersed Repetitive Sequences

Long Interspersed Nuclear Elements (LINEs ~6000 base pairs):

- Found in large numbers in eukaryotic genomes.
 - They are able to make an RNA.
 - LINEs include a gene that encodes the enzyme reverse transcriptase
- Reverse transcriptase makes a DNA copy of the LINE or SINE mRNA that can be integrated into the genome at a new site.
- LINEs are therefore capable of copying themselves and may enlarge the genome. The human genome contains 100,000s of LINEs
- LINE and SINE sequence repeats may contribute to mutation by leading to unequal crossover during meiosis
 - (same idea as for Low Copy Repeat sequences)

Highly repeated Interspersed Repetitive Sequences

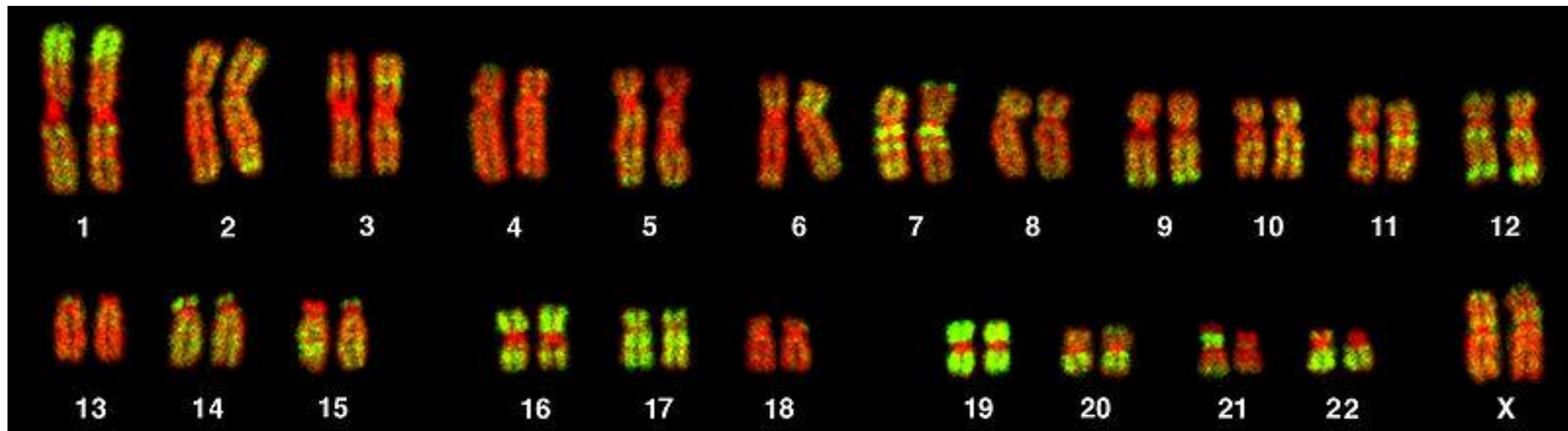
SINES and LINES (Make up almost half of the genome)

Short interspersed nuclear elements (SINEs)

- Short sequences of **<500 bp (maybe 300 bp)** that are found up to 1,500,000 times (millions) in the genome – approximately ~10% of the human genome

- Appear to be “normal” RNAs that were converted to DNA by reverse transcriptase and were reinserted into the genome. Reverse transcriptase is hijacked from a LINE.

- Most common SINE in humans is the *Alu* sequence. Called *Alu* because SINEs contain a sequence that is recognized by the Alu restriction enzyme.



Chromosomes hybridized with a Green probe for Alu sequences and DNA counterstained in Red

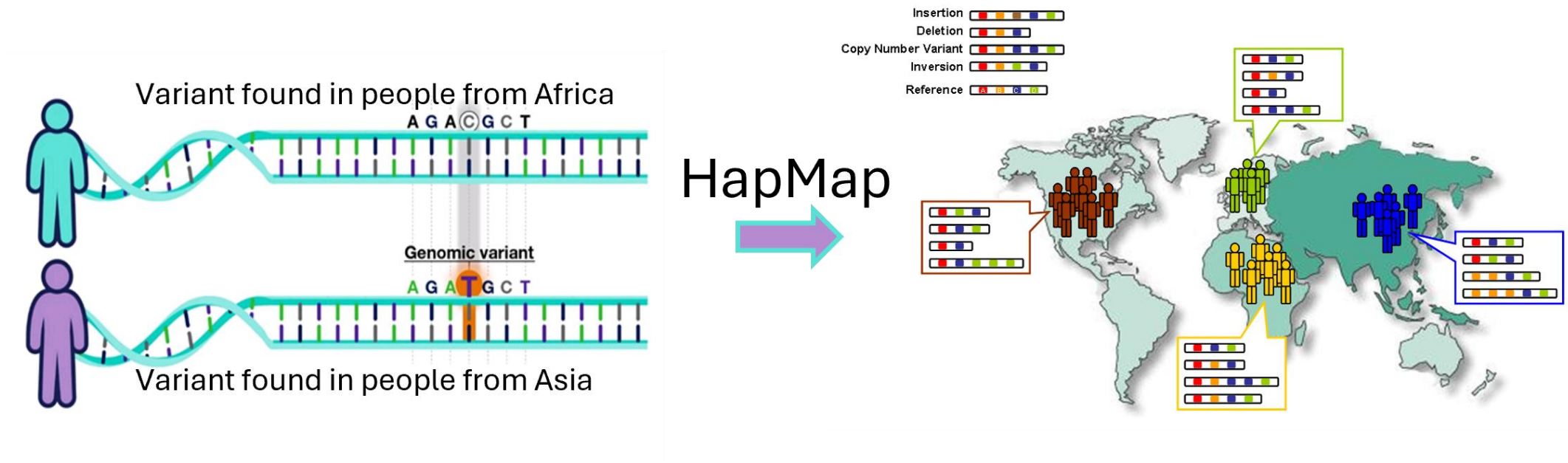
Bolzer et al., (2005) Three-Dimensional Maps of All Chromosomes in Human Male Fibroblast Nuclei and Prometaphase Rosettes. PLoS Biol 3(5): e157 DOI: 10.1371/journal.pbio.0030157, Figure 7a

The SNP Vs. the Rare variant

- SNPs are commonly found in many individuals (at least 1% of the population)
 - In total, there are only about 3 million or so of them
 - They are identified
 - So there are not many of them, but they are common
 - Thought to have arisen early in human evolution
- Rare variants are rare in the population... but, may be found in any of the 3 billion base pairs of the haploid genome
 - So rare variants are “*rare*” in that they are not frequently found in the population, but there may be billions of them
 - Another way of saying this: there are many more rare variants than there are common variants – they just aren’t frequently found
 - We do not know all of these rare variants, and there will always be a new one to discover
 - Rare variants are thought to contribute to human disease more than SNP

Stop and think about it point during study time: explain why there is virtually an unlimited amount of rare variants possible in our genome. Explain why most SNPs would not be causative of disease or a disorder

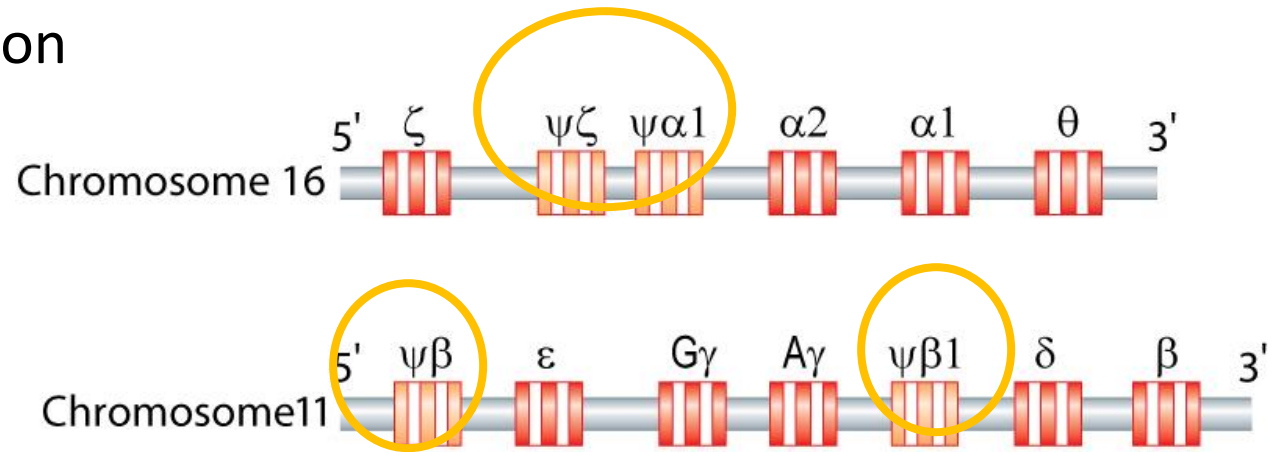
International HapMap Project



The **HapMap Project** aims to create a comprehensive catalog of common human genetic variations, specifically **single nucleotide polymorphisms (SNPs)** and their patterns of inheritance (**haplotypes**), across different populations. Its primary purpose was to provide a **public resource** enabling researchers to identify genetic variants associated with susceptibility to common diseases, drug responses, and other complex traits, thereby accelerating the discovery of genes affecting human health.

Pseudogenes

- Sequences that look like real genes but are not functional (**no protein product**). Most probably arose during evolution by:
 - Gene duplication and subsequent mutation
 - Copying of RNA back to DNA (by viral reverse transcriptases) and reinsertion into genome
 - The gene is turned off for some reason



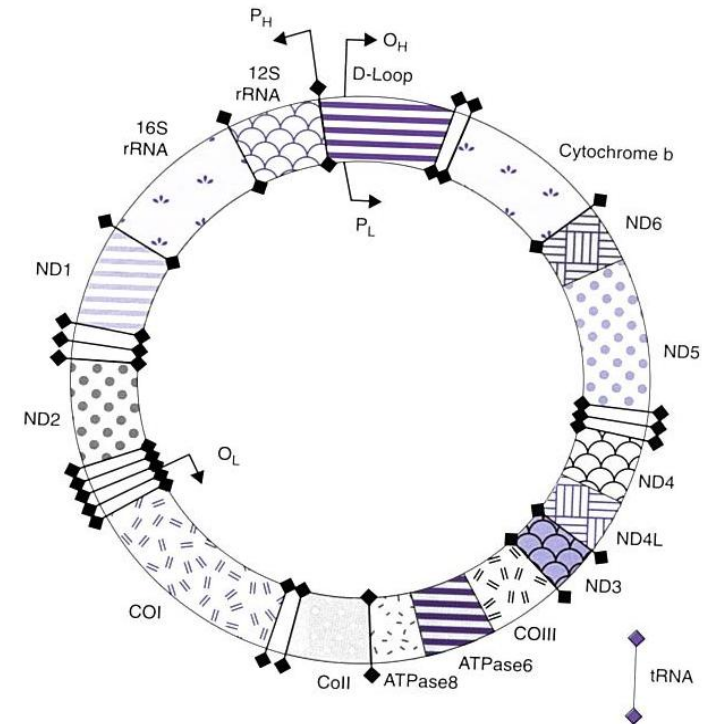
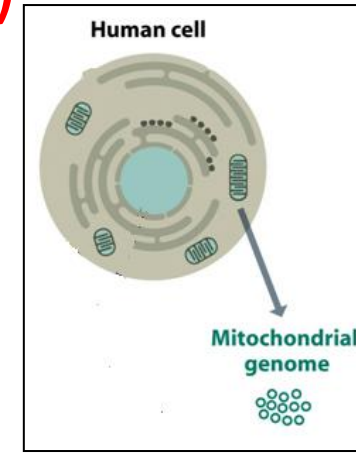
Expressed genes (in red) are transcribed & functional

Pseudogenes (in orange) are non-functional copies of a gene

- Conventional pseudogenes** (ψ) are not expressed & they often occur in gene families. Others can be expressed (transcribed) but their transcripts are not functional

Mitochondrial DNA (non-nuclear DNA)

- Mitochondria thought to have arisen by endosymbiosis.
- Single circular DNA of 16.6 Kb.
- Codes for some mitochondrial tRNA, and other factors needed for mito gene expression.
- Codes for a few mitochondria proteins
- Most mitochondrial proteins are encoded by genes in the nucleus.
- Maternally inherited in humans (all mitochondria come from the cytoplasm of the oocyte).
- Mitochondrial disorders are only passed through maternal line.
- Mitochondrial genes are more prone for mutation.

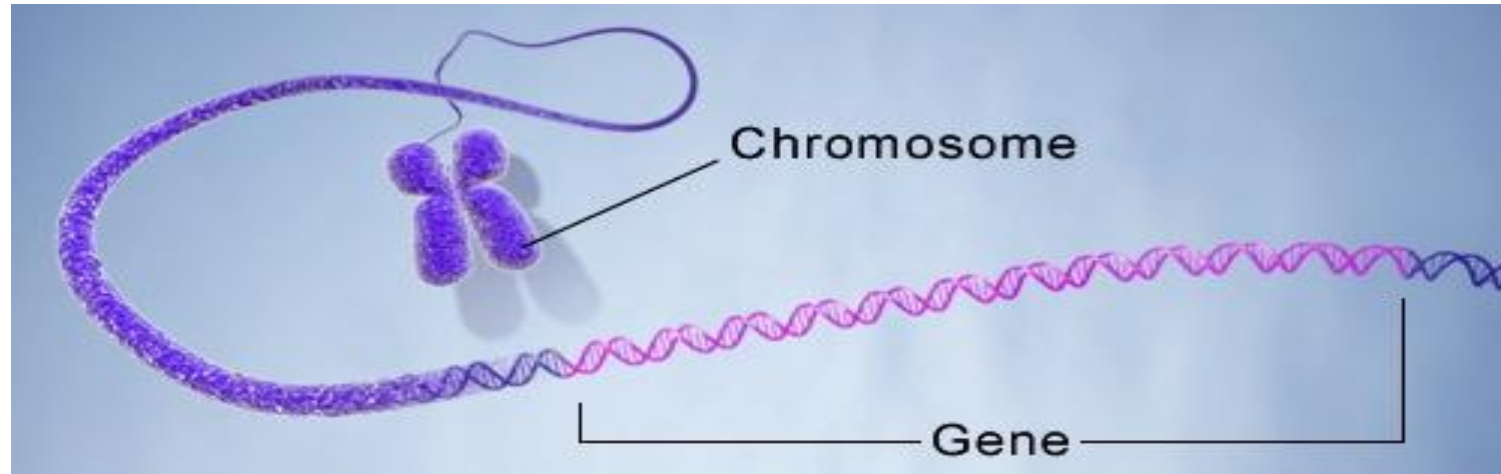


Map of the Mitochondrial Genome

(Korf p. 56)

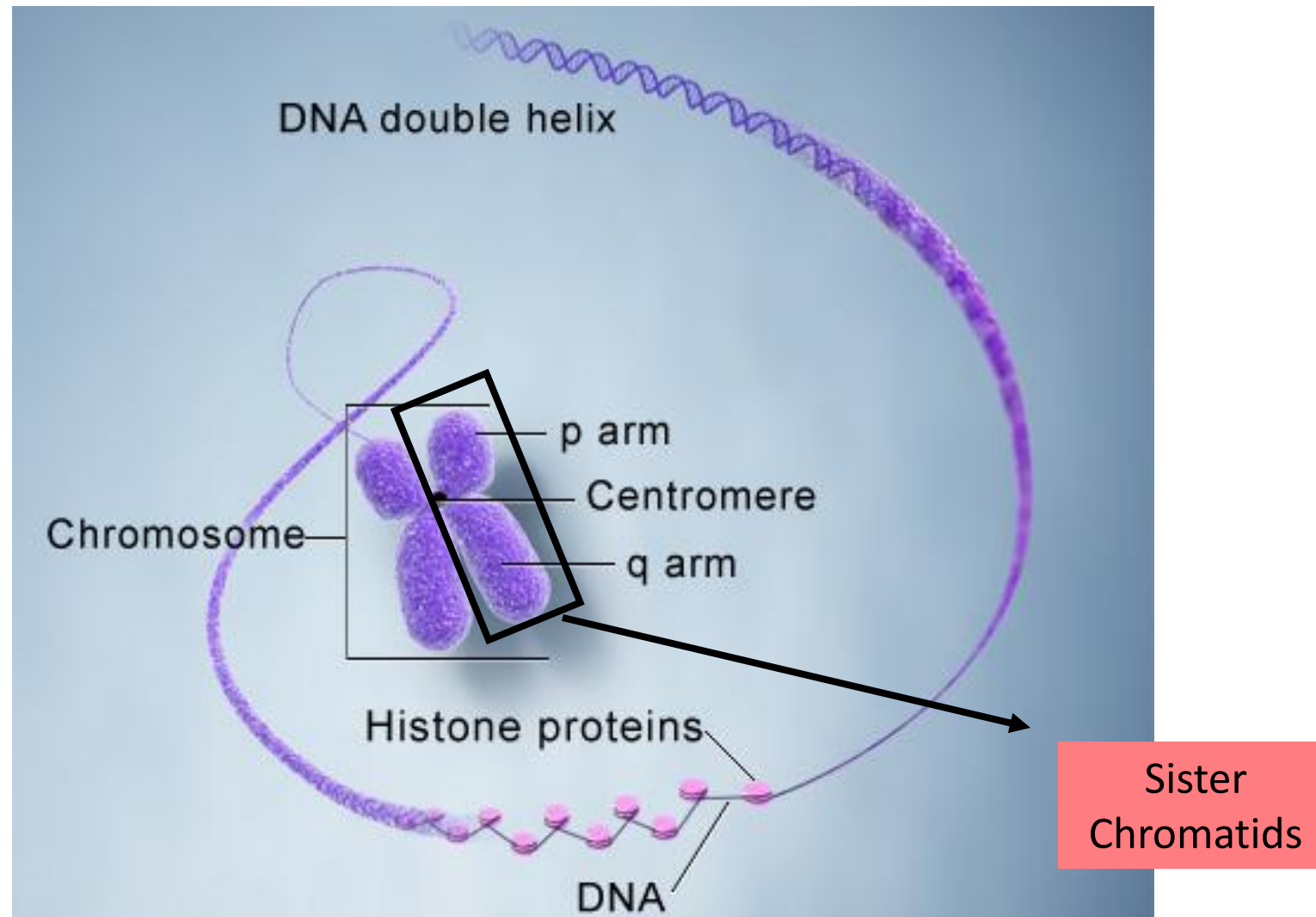
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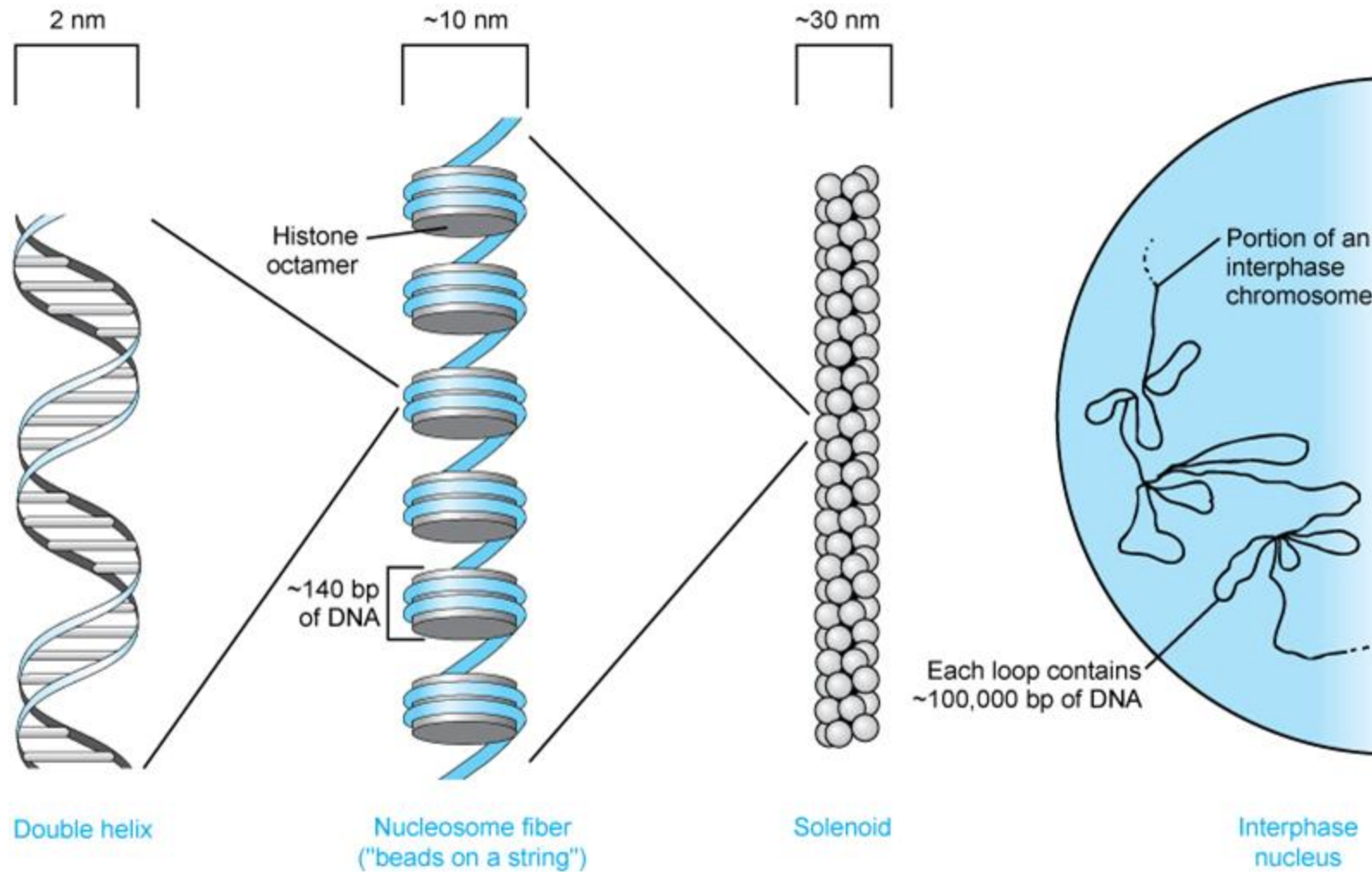
Chromosome



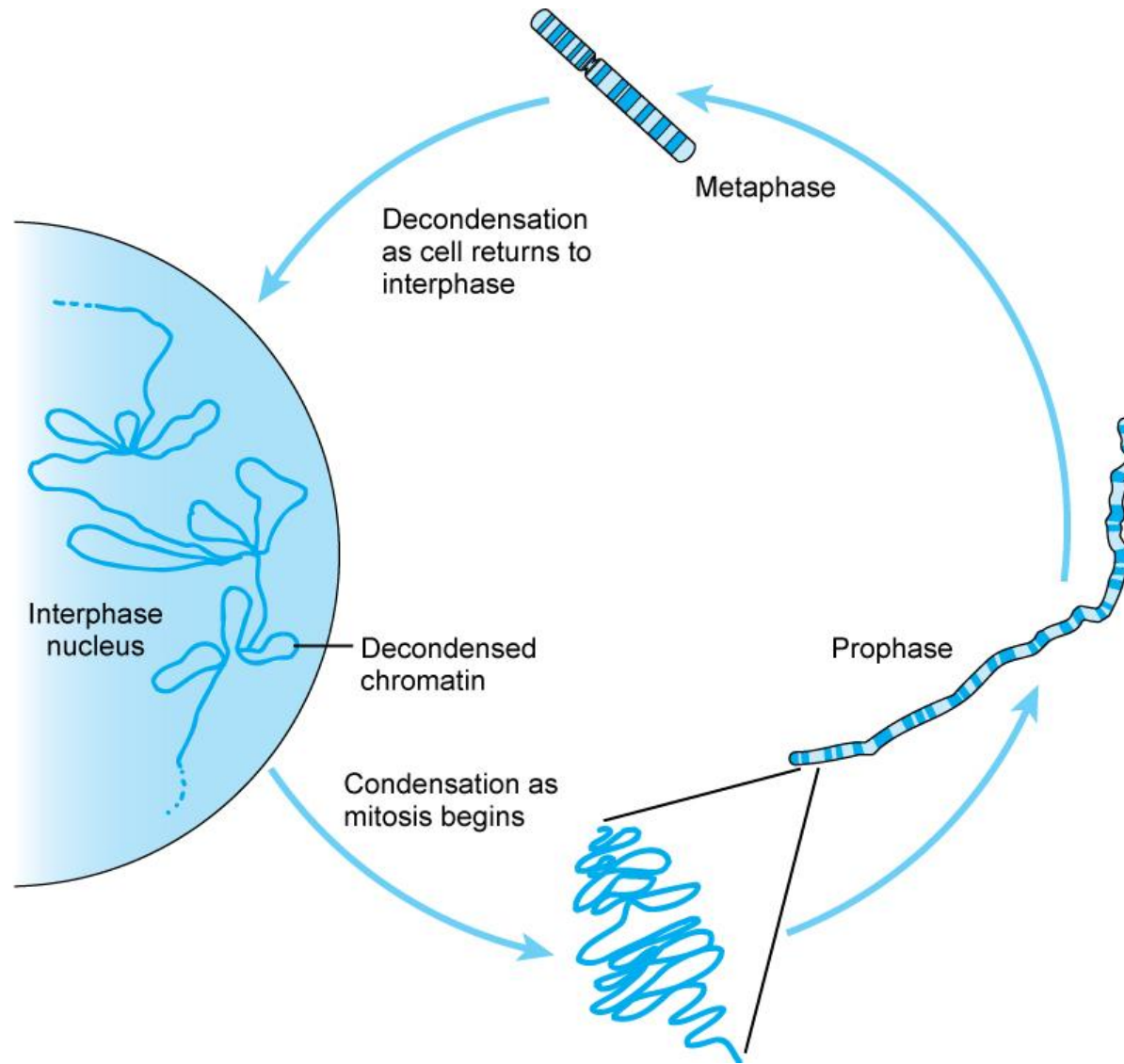
- Made up of a long chain of DNA
- The DNA is wound around specific (histone) proteins, to form a compact structure: Chromosome
- Each chromosome carries a different subset of genes that are arranged linearly along its DNA.
- Each chromosome is made up of many contiguous genes
- Chromosome has a constricted region: Centromere
- Centromere divides the chromosome into two arms:
 - p arm (short arm) and q arm (long arm)

Chromosome during metaphase





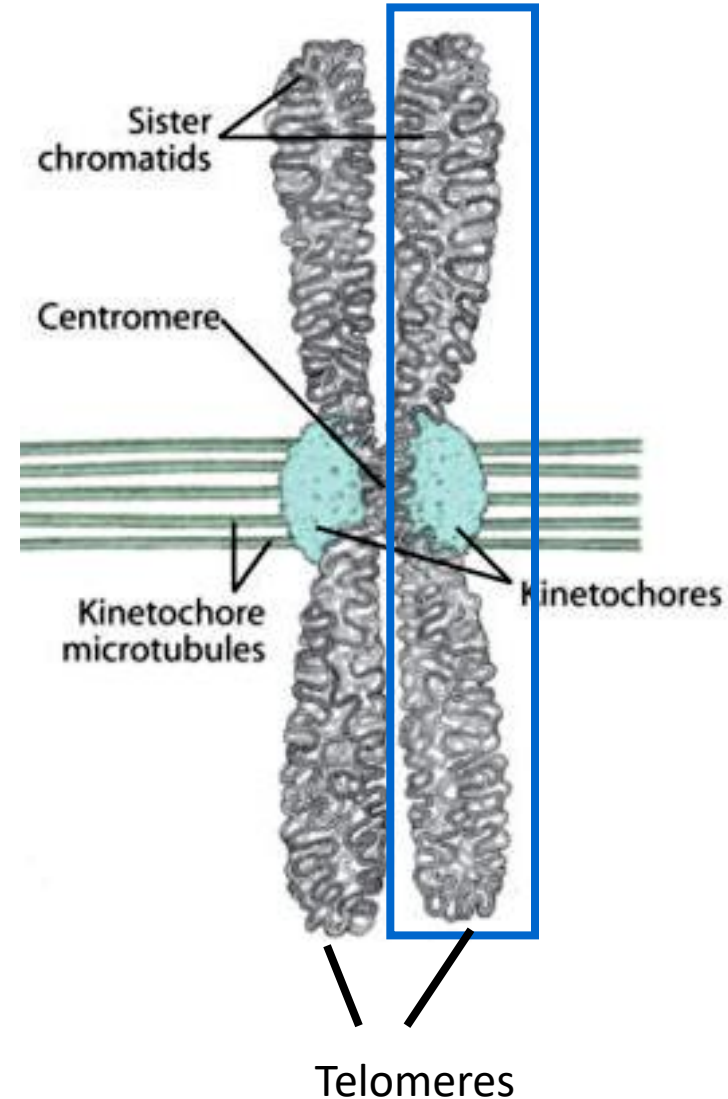
Hierarchical levels of chromatin packaging in a human chromosome



Cycle of condensation and decondensation as a chromosome proceeds through the cell cycle

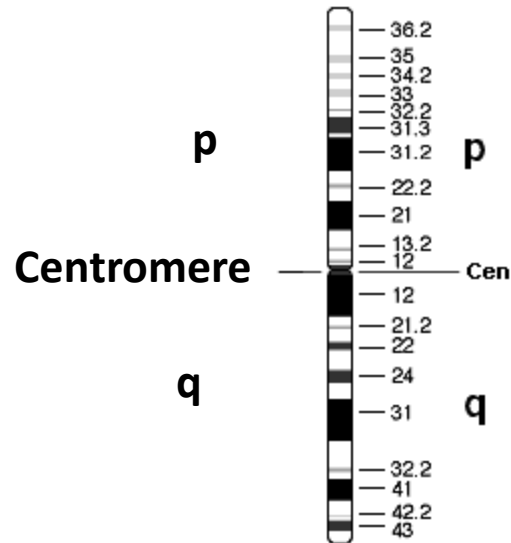
Chromosome at Metaphase

- Appearance of a chromosome during metaphase when they are maximally condensed and most visible.
- The microtubules attach to the centromere kinetochores during cell division.
- The ends of a chromosome are 'Telomeres'.
- Genes are usually absent in the telomeres and the centromere.



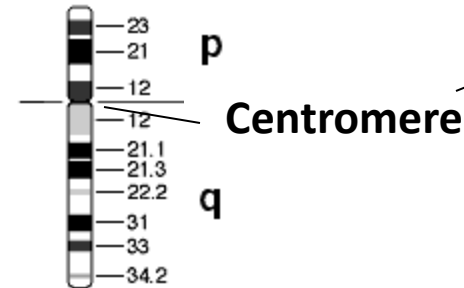
Chromosome Nomenclature

Metacentric chromosome



p and q arms are of equal length. Centromere in the centre: **Chromosome 1**

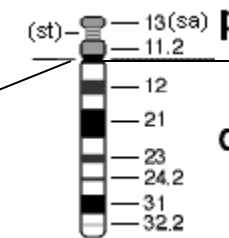
Submetacentric chromosome



p arm is shorter than q arm: **Chromosome 4**

Acrocentric chromosome

Satellite structures

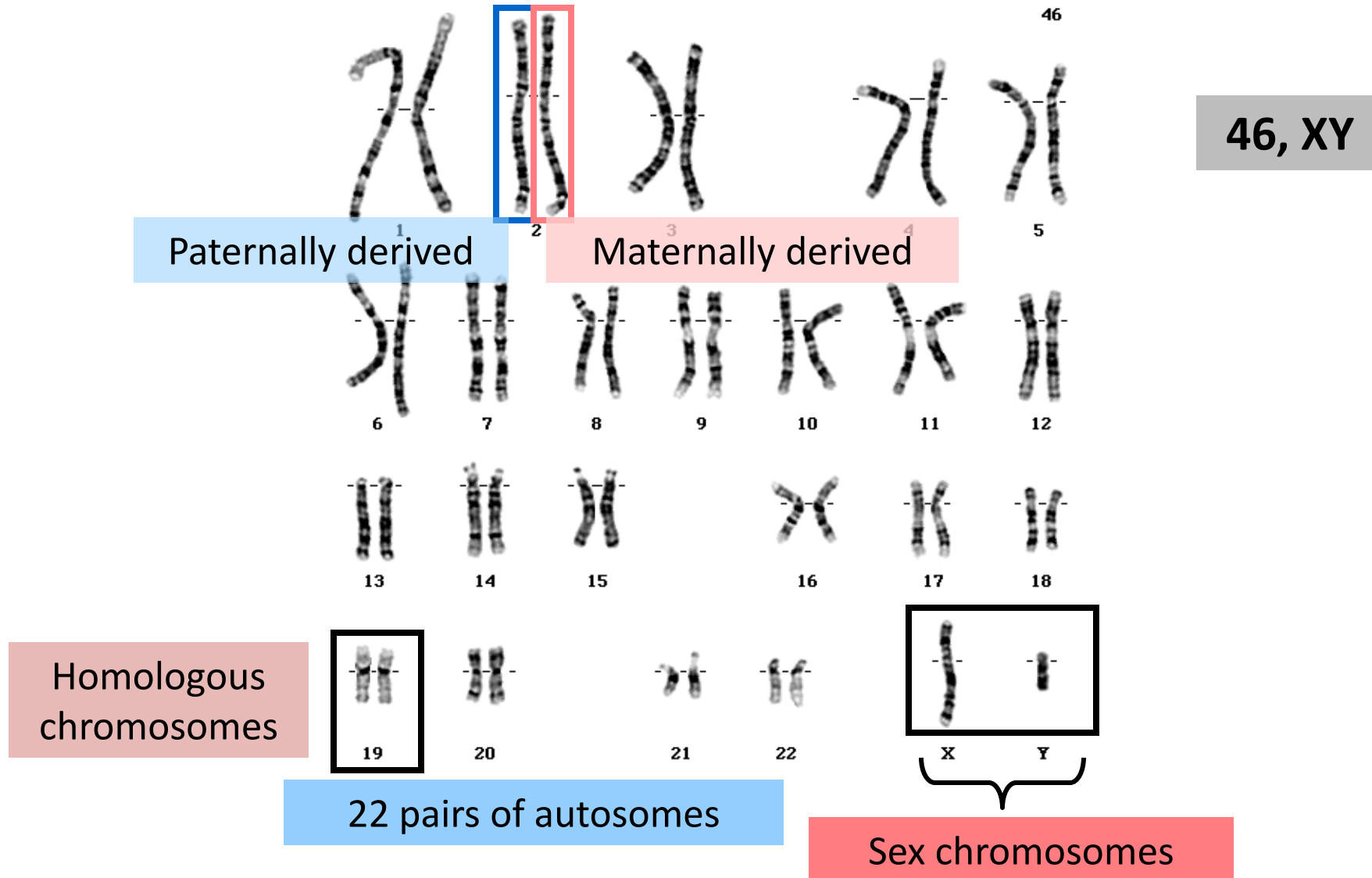


p arm contains little genetic information.
Chromosome 13, 14, 15, 21, 22: Involved in Robertsonian translocation

Karyotyping

- Karyotype: Visualization of chromosomes during **metaphase** (maximally condensed)
- Stained with specific stains to form a banded pattern, that is specific for a chromosome (G-banded karyotype)
- 22 pairs of autosomes and one pair of sex chromosomes (**Total: 23 pairs of chromosomes**)
- Chromosomes are ordered according to size: Chromosome 1 is the largest and Ch 21 is the smallest of the chromosomes
- One chromosome of the pair is derived from the mother and the other chromosome is derived from the father (when both the chromosomes of the pair are derived from the same parent, it is said to be **uniparental disomy - UPD**).

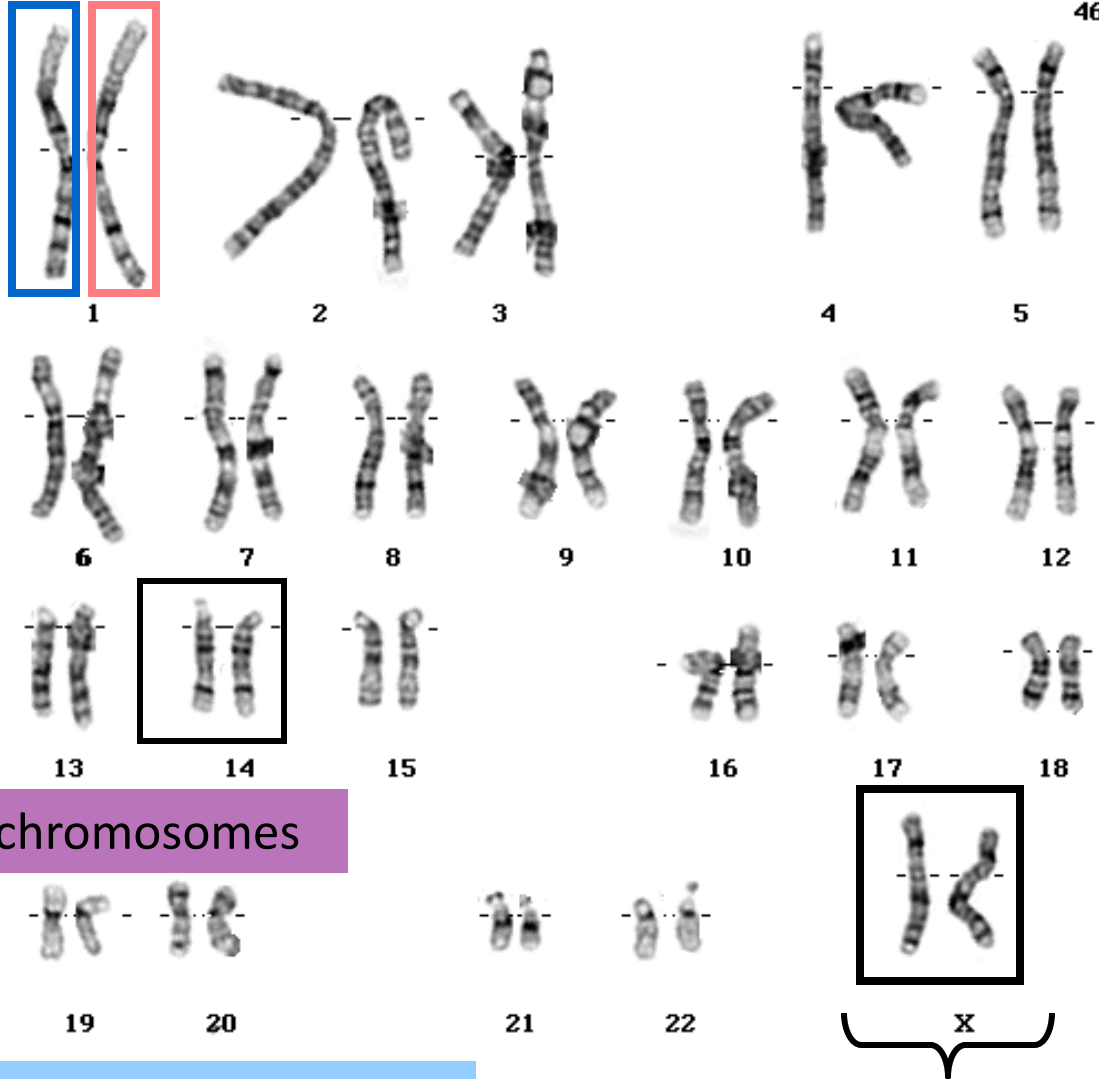
Normal Karyotype - Male



Normal Karyotype - Female

Paternally derived

Maternally derived



46, XX

Homologous chromosomes

22 pairs of autosomes

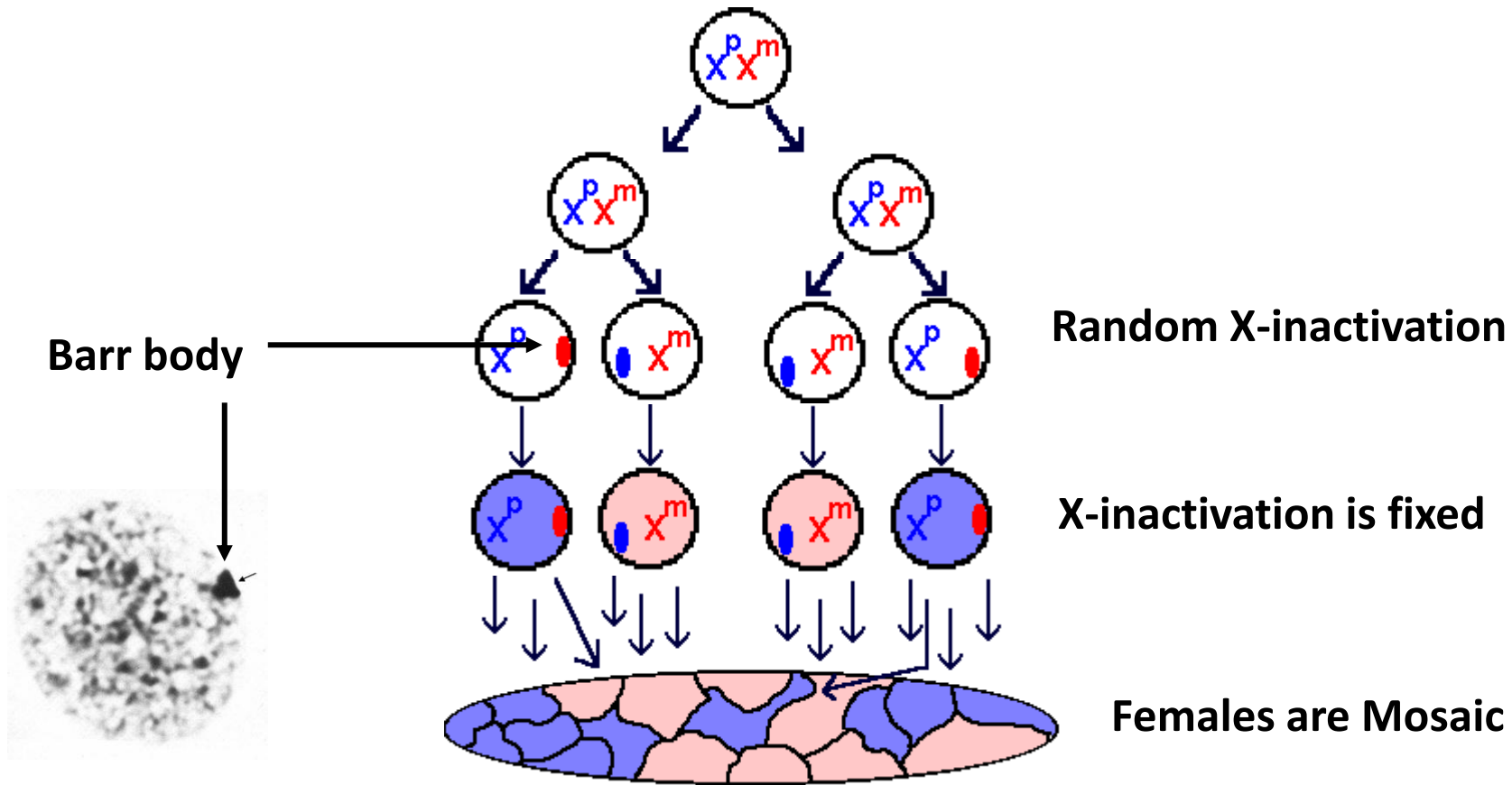
Sex chromosomes

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X-inactivation (Lyonization)

- Occurs during the early stages of development of female embryos
- One of the X-chromosome is inactivated and condensed to form the **Barr body** in females
- X-inactivation is **random** (Paternal X chromosome active in approximately 50% of cells and maternal X chromosome is active in approximately 50% of cells) ([Asymmetric X-inactivation](#) is a phenomenon observed when in more than 50% of the cells the paternal/maternal X is active)
- X-inactivation is **fixed**: The same X chromosome is inactivated in all the descendants of the cell
- X-inactivation is regulated by a region called **X-inactivation center (Xic)** that has the gene, **XIST gene** involved in inactivation
- XIST RNA coat one of the X-chromosomes, that results in transcriptional interference

X-inactivation



My Kali girl

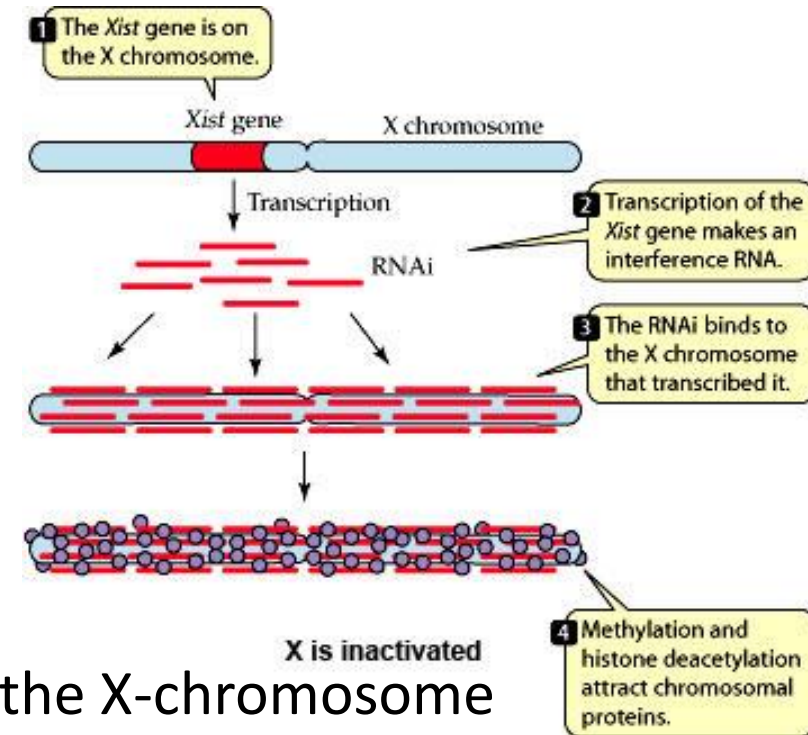
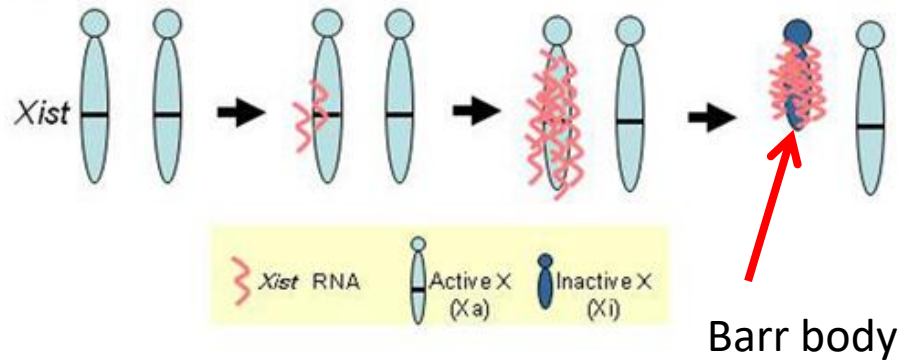
Paternal X chromosome active in approximately 50% of cells and maternal X chromosome is active in approximately 50% of cells

Pseudoautosomal regions of the X chromosome are not inactivated

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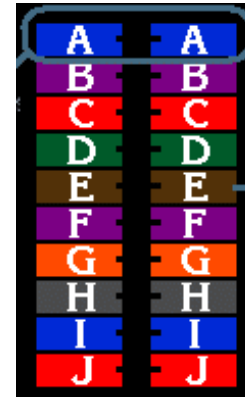
X-inactivation



- *Xist* gene produces an RNA that coats the X-chromosome
- Causes X chromosome condensation to heterochromatin
 - cytosine methylation of specific regions occurs (epigenetic mechanism).
 - Only specific regions of the X chromosome are methylated.
 - Some of the X chromosome escapes inactivation
- Methylation of cytosine bases (CG islands) and histone deacetylation, result in transcriptional repression
- The process of imprinting is similar to X-inactivation, in that the imprinted gene is methylated and transcriptionally silenced.

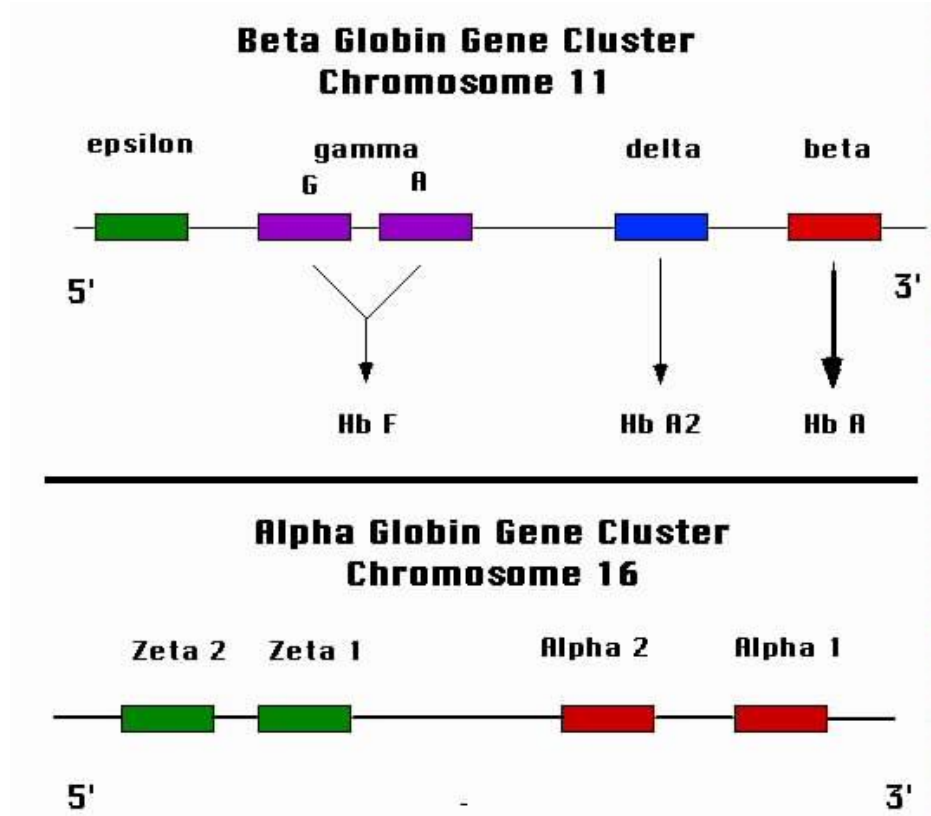
Alleles of a Gene

- Members of a pair of chromosomes (**homologous chromosomes**) carry matching genetic information:



- Same genes in the same order. At any specific locus, however, they may have either **identical (homozygous)** or slightly **different (heterozygous)** forms of the same gene, called alleles.
- Alleles are different versions of the same gene; they have slight differences in the base pair DNA sequence
- If the homologous chromosomes have the same allele of a gene, then the person is **homozygous** for that allele
- If the homologous chromosomes have different allelic forms of a gene, then the person is **heterozygous** for that allele

Locus of a gene



- **Locus** of a gene refers to its location on a chromosome
- The Huntingtin gene locus is on chromosome 4p
- Beta globin gene cluster is on chromosome 11
- Alpha globin gene on chromosome 16



In most cases, we do not ask you to memorize the locus of a gene. Notable exceptions will be made clear; for instance, X-linked disorders, and a few translocation events

Mendelian Inheritance in Man (MIM)

- Victor McKusick and colleagues at Johns Hopkins University developed a cataloging system for human genetic traits and genes
- Began as a library project (JHU medical library)
- Advances outpaced ability to update library entries, so online Mendelian Inheritance in Man (OMIM) project created.
 - Now maintained by NCBI
- Each genetic trait has a unique six-digit MIM number

MIM number

- Autosomal dominant disorders begin with 1
 - Example: Marfan syndrome (MIM 154700)
- Autosomal recessive disorders begin with 2
 - Example: Cystic fibrosis (MIM 219700)
- X-linked disorders begin with 3
 - Example: Duchenne muscular dystrophy (MIM 310200)
- Mitochondrial disorders begin with 5
 - Example: Leber hereditary optic neuropathy (MIM 535000)

We do not expect you to memorize these cataloging numbers

Specific points for students to think about while studying

- Summarize how DNA is able to act as the store of genetic information
- Describe the structure of normal human chromosomes; Analyze chromatin structure
- Define what constitutes a normal karyotype
- Define, then compare and contrast the terms: 'gene' 'alleles of a gene', 'homologous chromosomes', 'homozygous', 'heterozygous', and 'locus'
- Describe the structure and organization of genes
- Discuss mRNA splicing and how alternative splicing increases gene diversity
Explain the potential consequences of the splice site mutation
- Provide a basic description of the structure of the human genome
 - Describe single gene sequences and unique sequences
 - Multigene families and the importance of pseudogenes
 - Discuss the distribution of repetitive sequences in the genome; Compare contrast genomic architecture and repetitive DNA
 - List and give describes examples: SSR's (simple sequence repeats), segmental repeats, pseudogenes, and transposons as types of repetitive sequences
 - Define the terms LINES and SINES and explain why SINEs are also known as the *Alu* sequence
- Discuss the X chromosome
 - Analyze the structure and function of the normal X-chromosome
 - Analyze X-chromosome inactivation
- Discuss some mechanisms the uses to pack approximately 6 meters of DNA into the nucleus which is only 5 μm in diameter
 - Describe the histone octamer and its importance in formation of the nucleosome
 - Describe the 30 nm solenoid fiber, and packaging of DNA into loops, formation of chromatin loops, and condensation of the metaphase chromosome
- Describe the general structure of the metaphase chromosomes including the terms: sister chromatid, centromere, telomere, microtubule, and kinetichore
- Recognize that there is a basis for cataloguing genetic disorders in OMIM (MIM number) – *do not memorize* the reference numbers used in OMIM.